Adrenal Pathology Lab
Block 7 Week 1

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UIC UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Conflict of Interest Statement

The presenters do not have a financial interest, arrangement, or affiliation with any organization that could be perceived as a real or apparent conflict of interest in the context of the subject of this presentation.

Try not to look ahead to the answers!

Answers and explanations are included in your slides for ease of post-session review. Try not to look ahead at the answers during this session for the best learning experience.



When you see this icon on the slide, it means the answer is on the next slide.

Learning Objectives

1. Describe the pathology of Cushing syndrome, Addison's disease, and Waterhouse-Friderichsen syndrome.
2. Compare and contrast adrenal cortical hyperplasia, adenoma and carcinoma.
3. Describe the clinical features, morphology, laboratory findings, treatment and prognosis of adrenal medullary neoplasms (pheochromocytoma and neuroblastoma).
4. Compare and contrast pheochromocytoma and paraganglioma.
5. Describe multiple endocrine neoplasia (MEN) syndromes in terms of clinical, laboratory, and morphologic features as well as prognosis.

Part 1: Adrenal cortical disorders

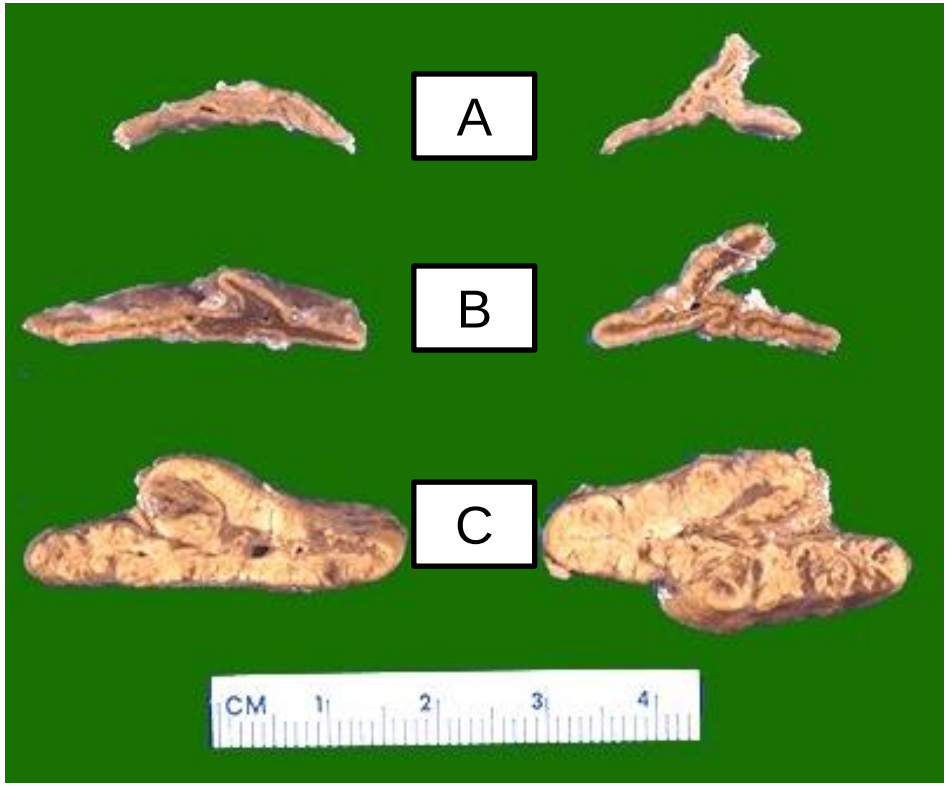
Case 1

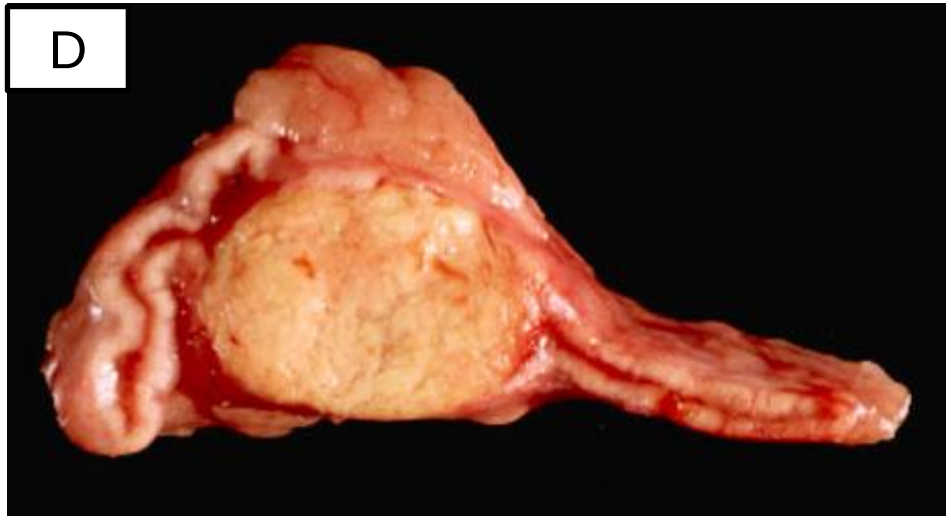


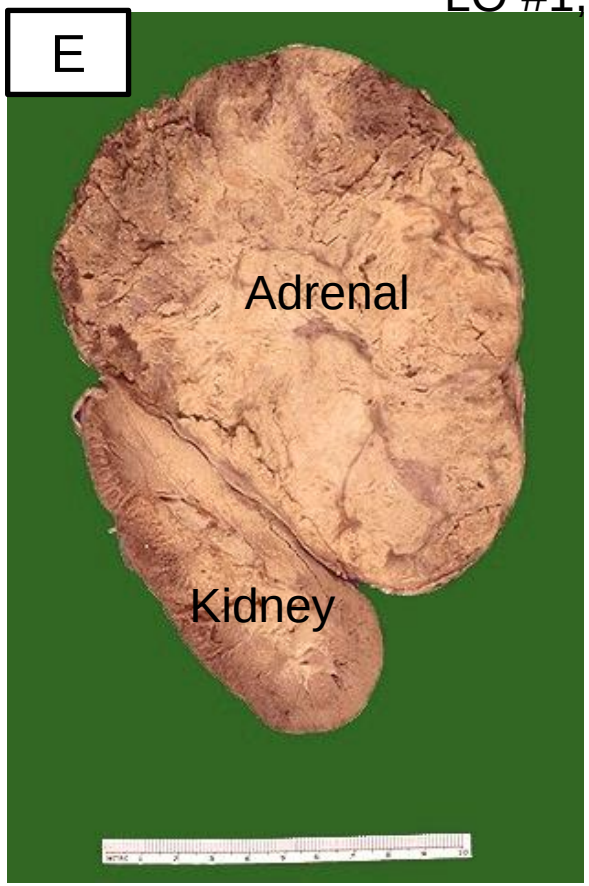
A 35-year-old man presents with proximal muscle weakness, easy bruisability and a loss of 2 inches in height due to having suffered a vertebral fracture. Recently, he was diagnosed with diabetes mellitus. He is currently taking metformin 500 mg twice daily.

What is the most likely pathologic change in the adrenal glands if…

1. He were on corticosteroids for ulcerative colitis?
2. His labs showed: elevated 24 hr urinary free cortisol and elevated plasma ACTH?
3. His labs showed elevated 24 hr urinary free cortisol and low plasma ACTH?

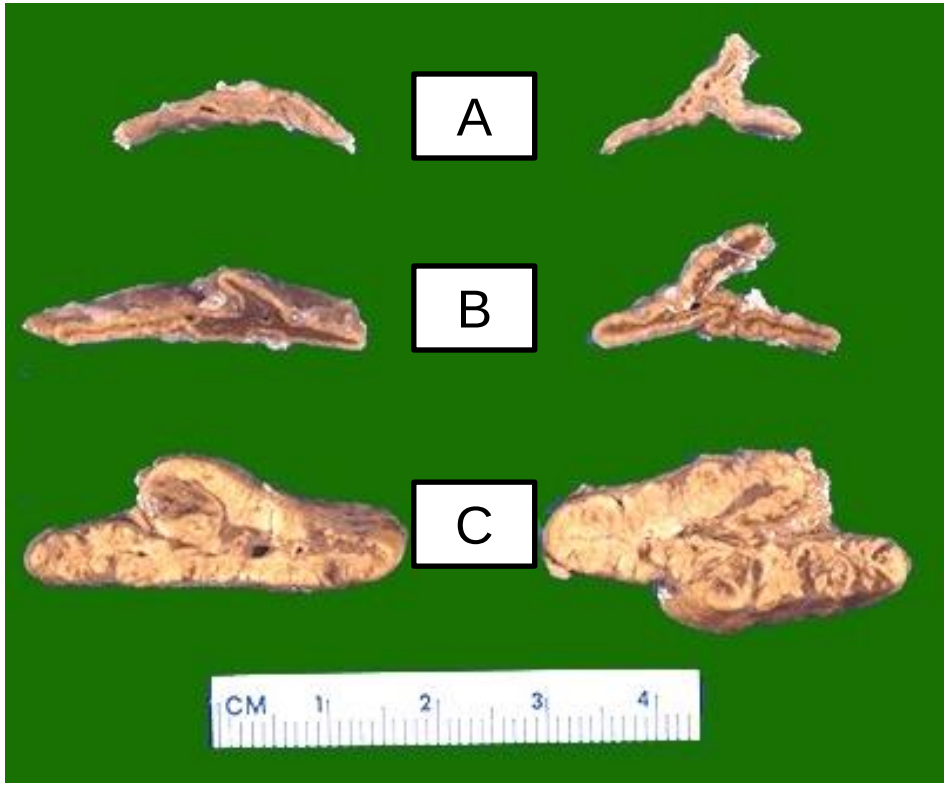


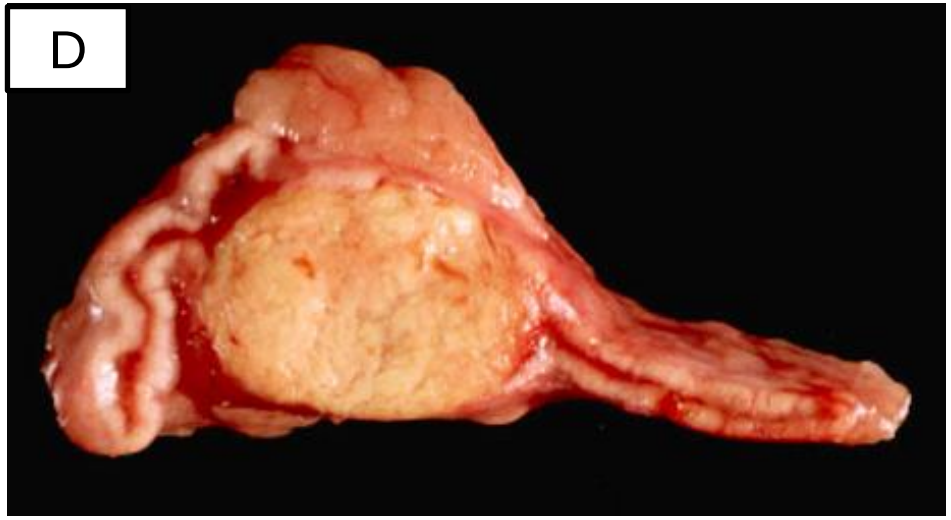


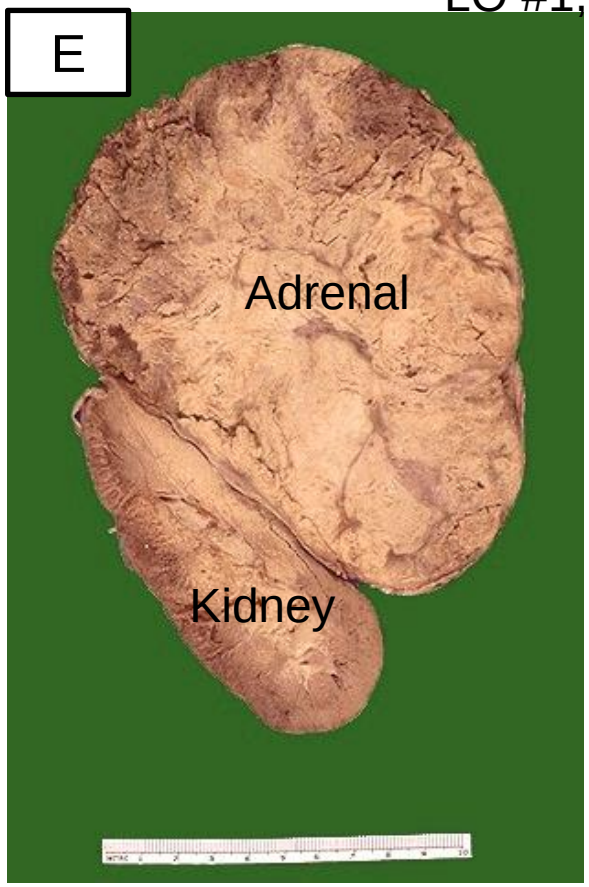


Note: B shows normal adrenal glands

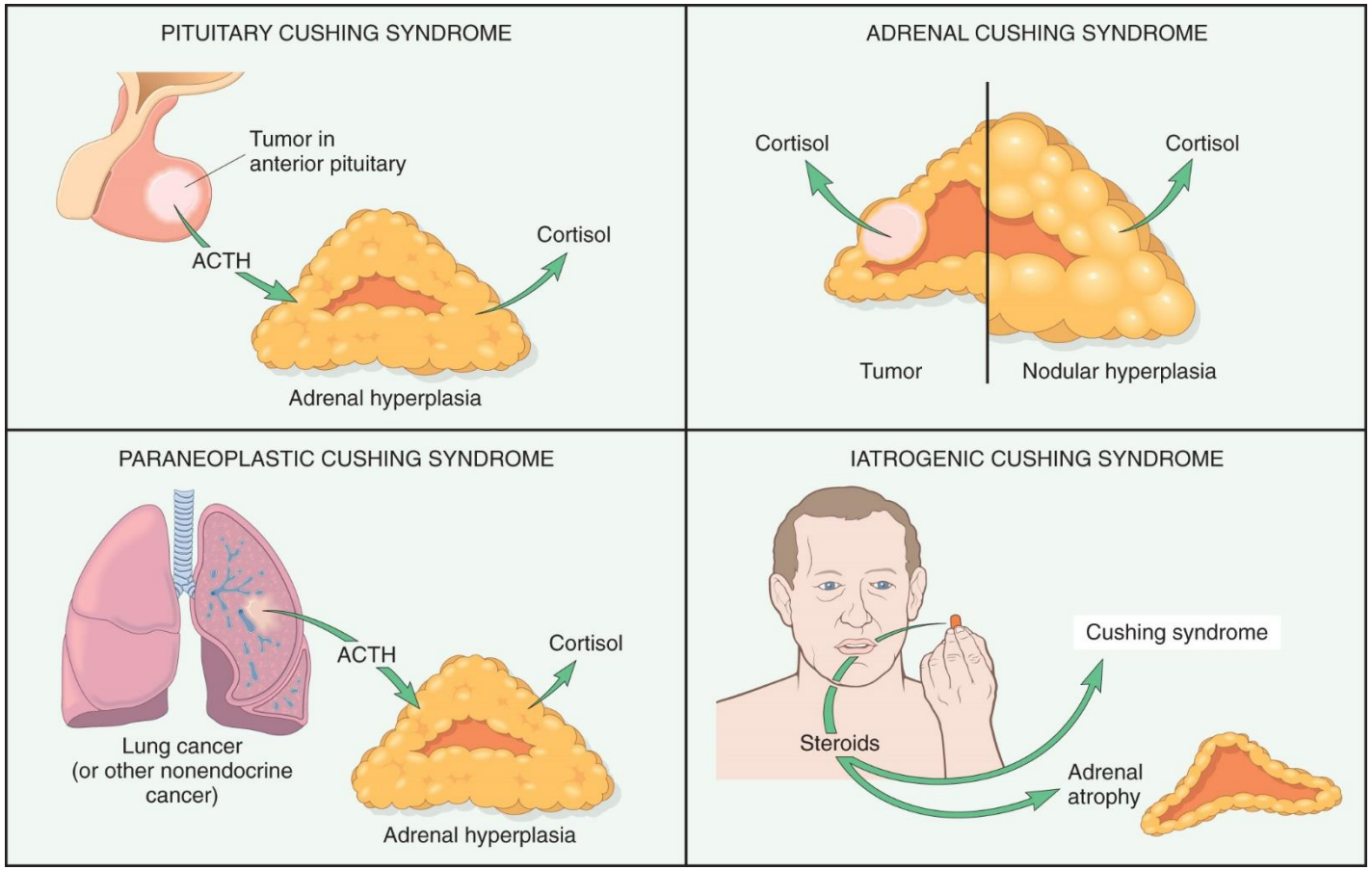
1. He were on corticosteroids for ulcerative colitis? **A**
2. His labs showed: elevated 24 hr urinary free cortisol and elevated plasma ACTH? **C**
3. His labs showed elevated 24 hr urinary free cortisol and low plasma ACTH? **D, E**







Note: B shows normal adrenal glands



Frequency of causes of Cushing's syndrome

Diagnosis	Percent of patients
ACTH-dependent Cushing's syndrome	
Cushing's disease	68
Ectopic ACTH syndrome	12
Ectopic CRH syndrome	<<1
ACTH-independent Cushing's syndrome	
Adrenal adenoma	10
Adrenal carcinoma	8
Micronodular hyperplasia	<1
Macronodular hyperplasia	<1
Pseudo-Cushing's syndrome	
Major depressive disorder	1
Alcoholism	<<1

Relative prevalence of various causes of Cushing's syndrome in 630 patients (146 consecutive patients seen at Vanderbilt University Medical Center before 1993 and published reports describing 484 patients). The prevalence of pseudo-Cushing's syndrome depends upon the individual clinician's threshold of clinical suspicion; in our experience, it is very rare. The relative prevalence of various causes of Cushing's syndrome among children and adolescents may differ somewhat from that of adults. The ectopic ACTH syndrome, for example, is less common in children.

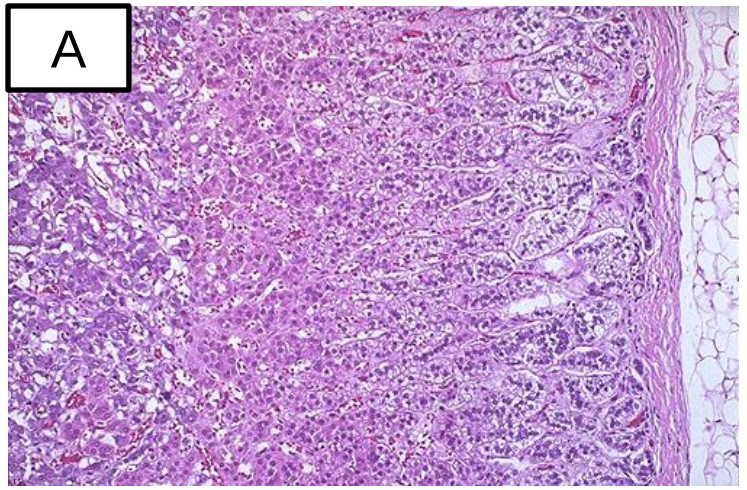
ACTH: corticotropin; CRH: corticotropin-releasing hormone.

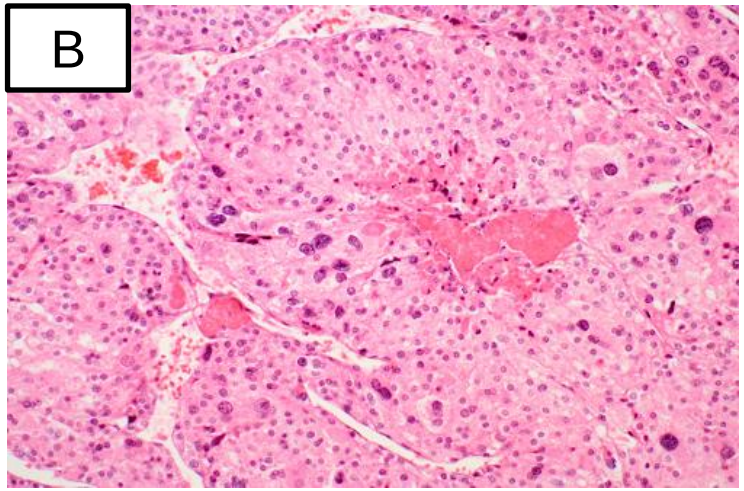
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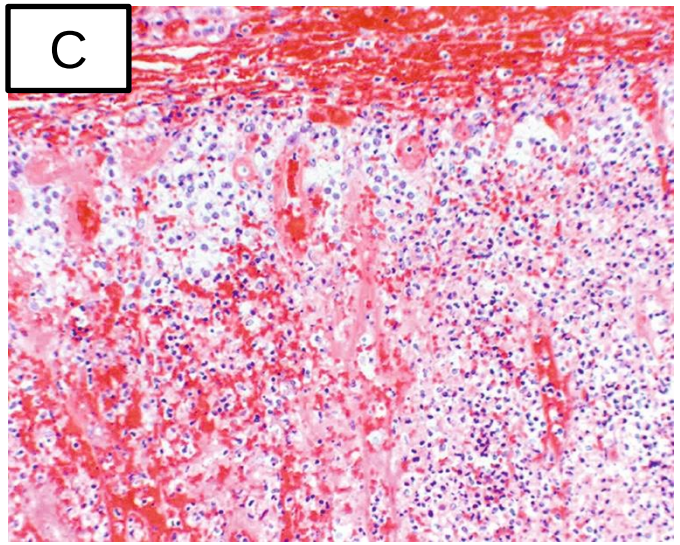
Case 2

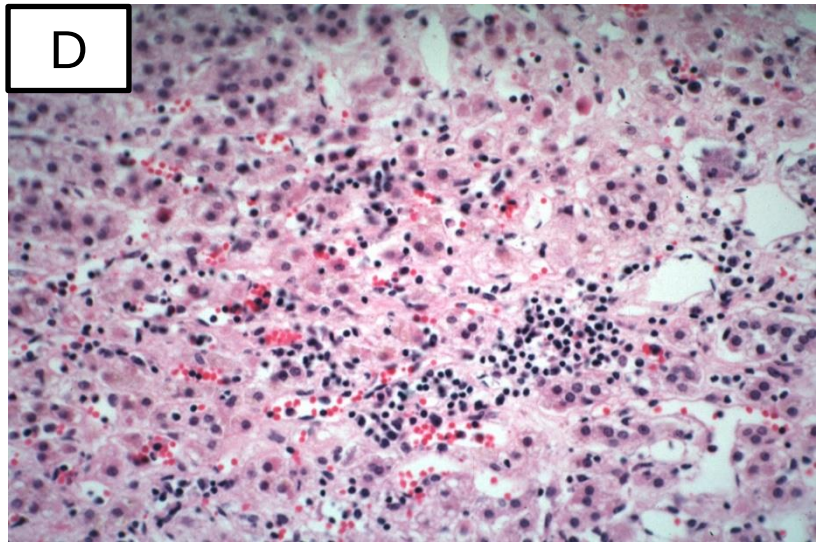
A 14-year-old girl is brought to the emergency department for high fever, chills, weakness, vomiting and severe headache. Over the next few hours she developed a red rash on her arms and legs that spread over her body, including mucous membranes. She became hypotensive and required intubation. Gram stain of her CSF showed Gram negative diplococci.

What do you expect this patient's adrenal glands to show?





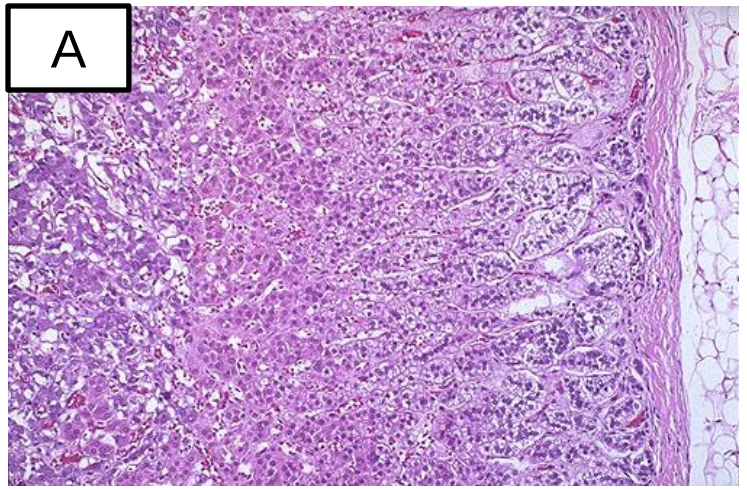


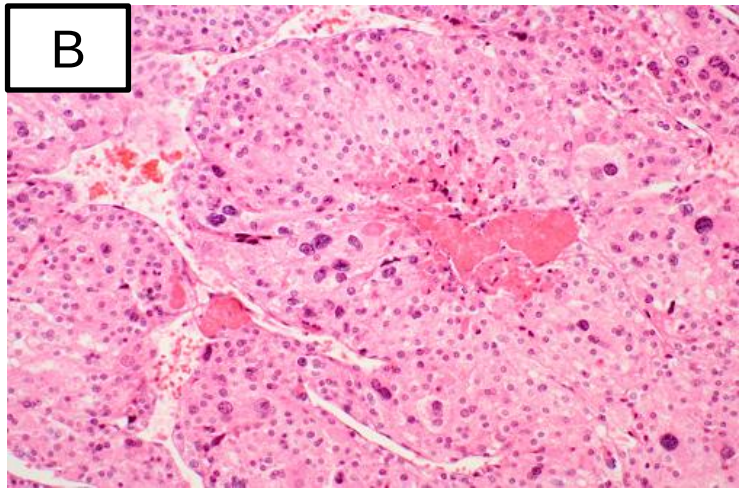


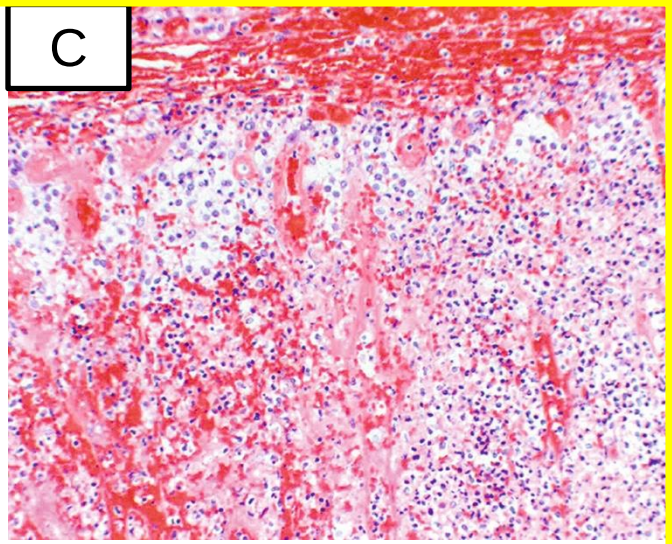
Answer: C

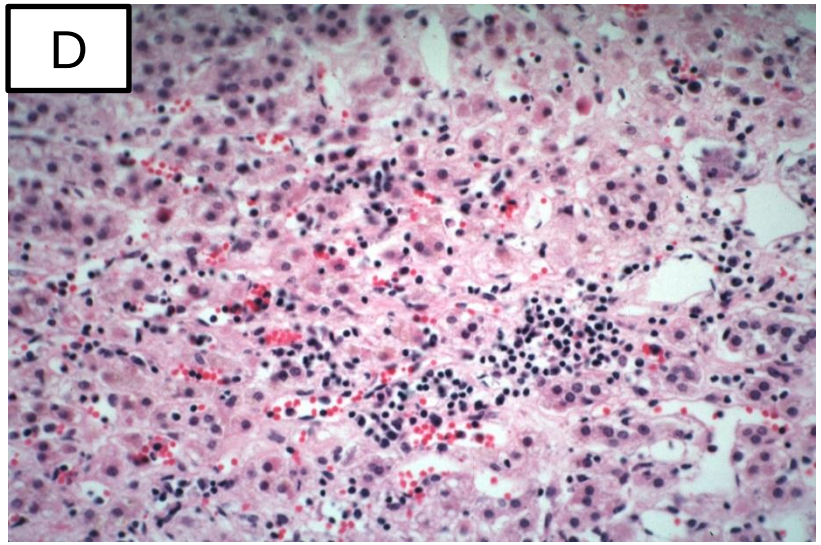
Diagnosis: Waterhouse-Friderichsen syndrome

What do you expect this patient's adrenal glands to show?



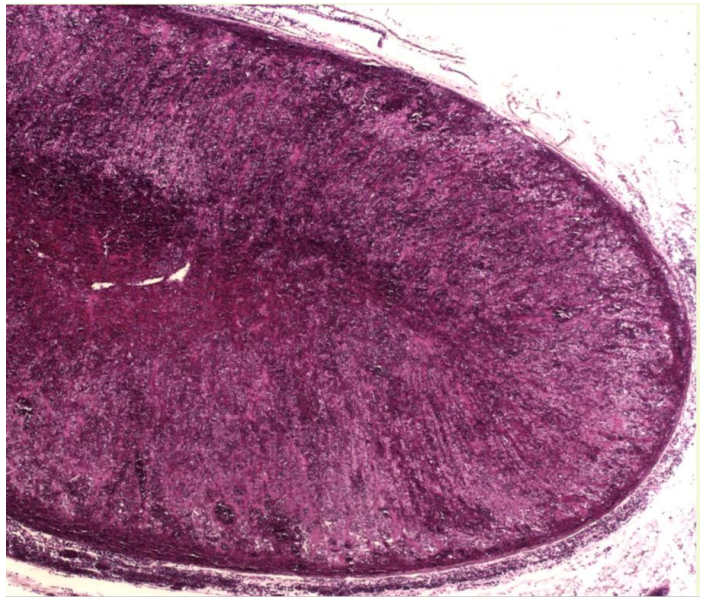


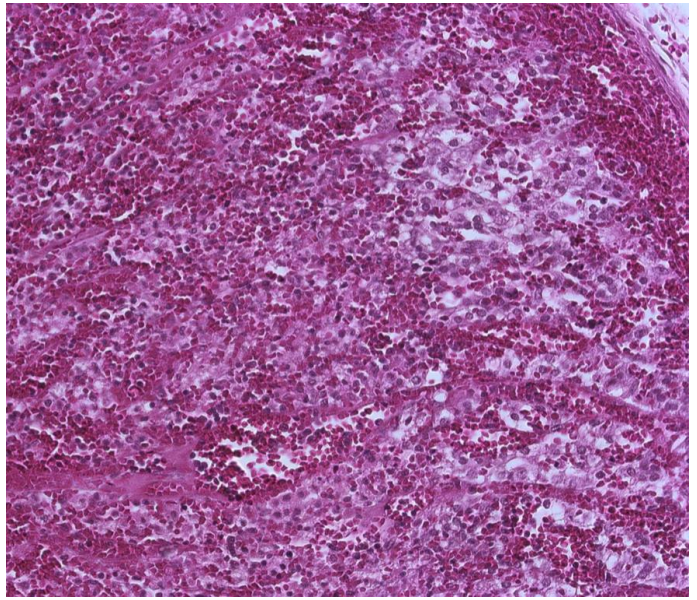




Acute Adrenocortical insufficiency due to Waterhouse- Friderichsen syndrome

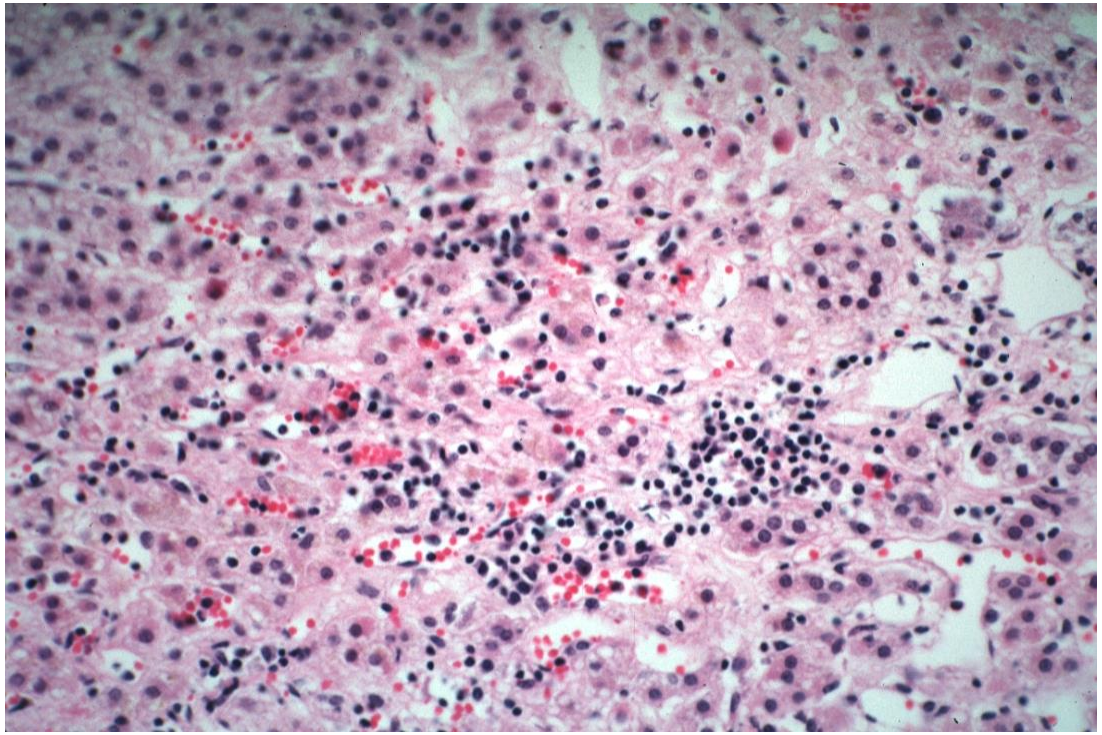






Chronic Adrenocortical insufficiency

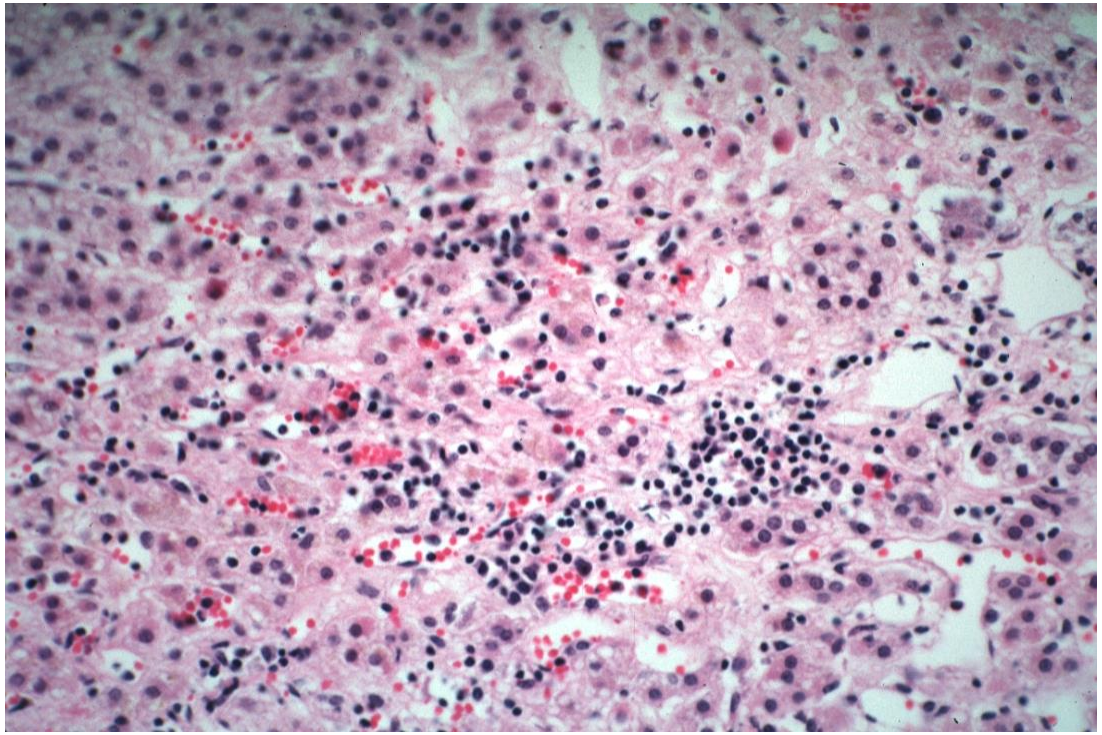




What are the most common causes of chronic adrenocortical insufficiency? (Name 4)

Chronic Adrenocortical insufficiency



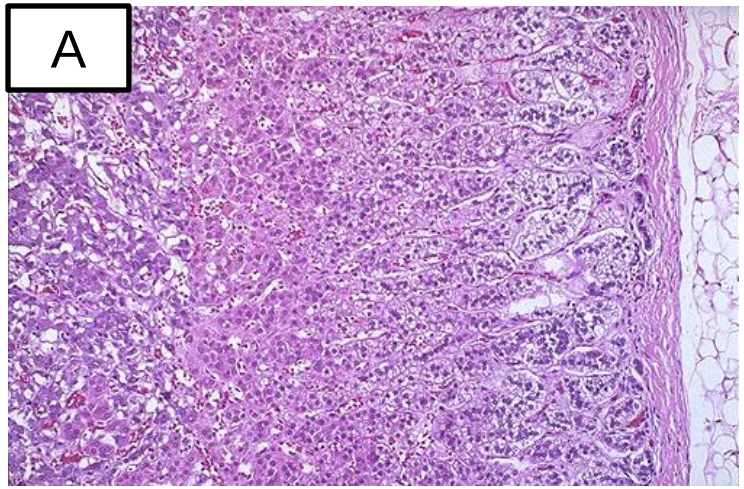


What are the most common causes of chronic adrenocortical insufficiency? (Name 4)

1. **Autoimmune adrenalitis (images, left)**
2. **Tuberculosis**
3. **AIDS (opportunistic infections, KS)**
4. **Metastatic cancer**

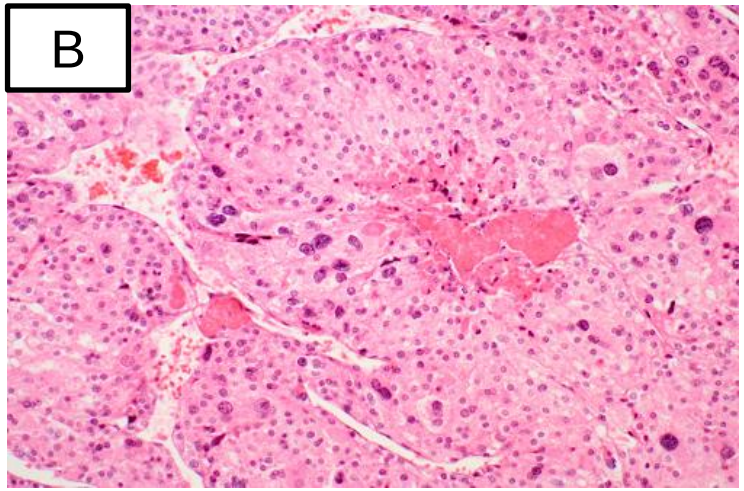
Explanation of other incorrect answers

A



- Normal adrenal gland

B



- Adrenal cortical carcinoma

Part 1: Summary

- Adrenal cortical hyperfunction: Cushing syndrome
- Adrenocortical hyperplasia vs. adenoma vs. carcinoma
- Adrenocortical insufficiency
  - Acute: Adrenal hemorrhagic necrosis (Waterhouse-Friderichsen syndrome)
  - Chronic: Addison disease

Part 2: Adrenal medullary tumors

Case 3

A 55-year-old woman undergoes an abdominal CT examination for evaluation of abdominal pain. The scan shows no specific abnormalities that correlate with abdominal pain, but a 3 cm incidental adrenal mass is identified in the right adrenal gland. On further questioning, she describes episodes during which she has felt sweaty, anxious and has had palpitations. During one of these episodes her blood pressure was 160/96 mm Hg.

Which of the following laboratory tests is the most sensitive to identify the biochemical basis for her symptoms?

A. Plasma metanephrine / normetanephrine
B. Plasma cortisol
C. Plasma ACTH
D. Plasma catecholamines
E. 24 hour urine 5 hydroxy-indole acetic acid (HIAA)

Answer: Case 3

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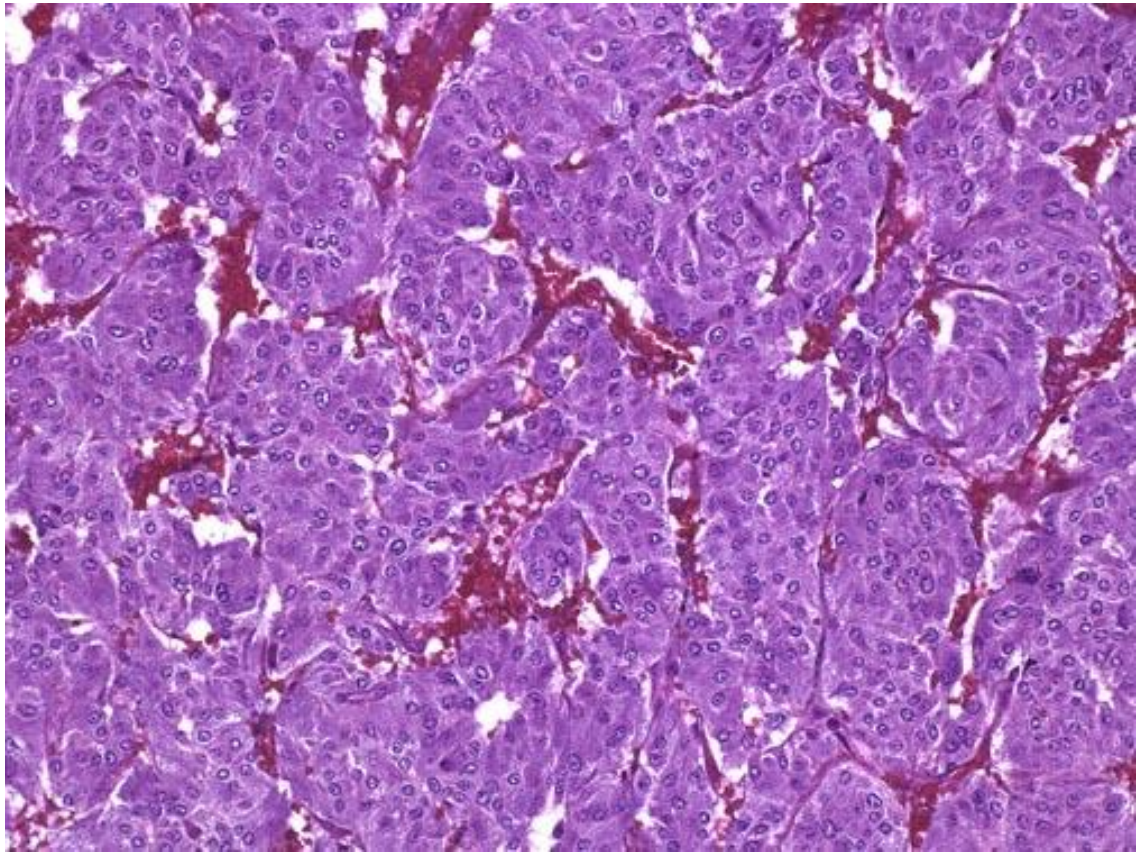
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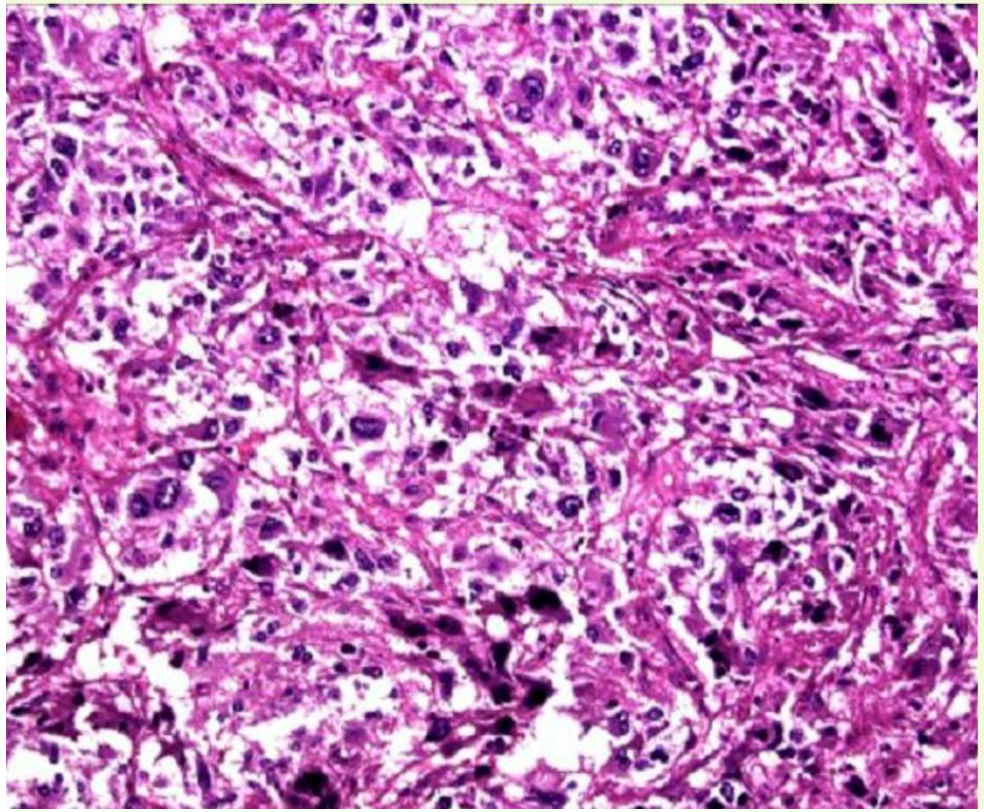
Pheochromocytoma
Gross and Microscopic features





Pheochromocytoma
Microscopic and Ultrastructural features

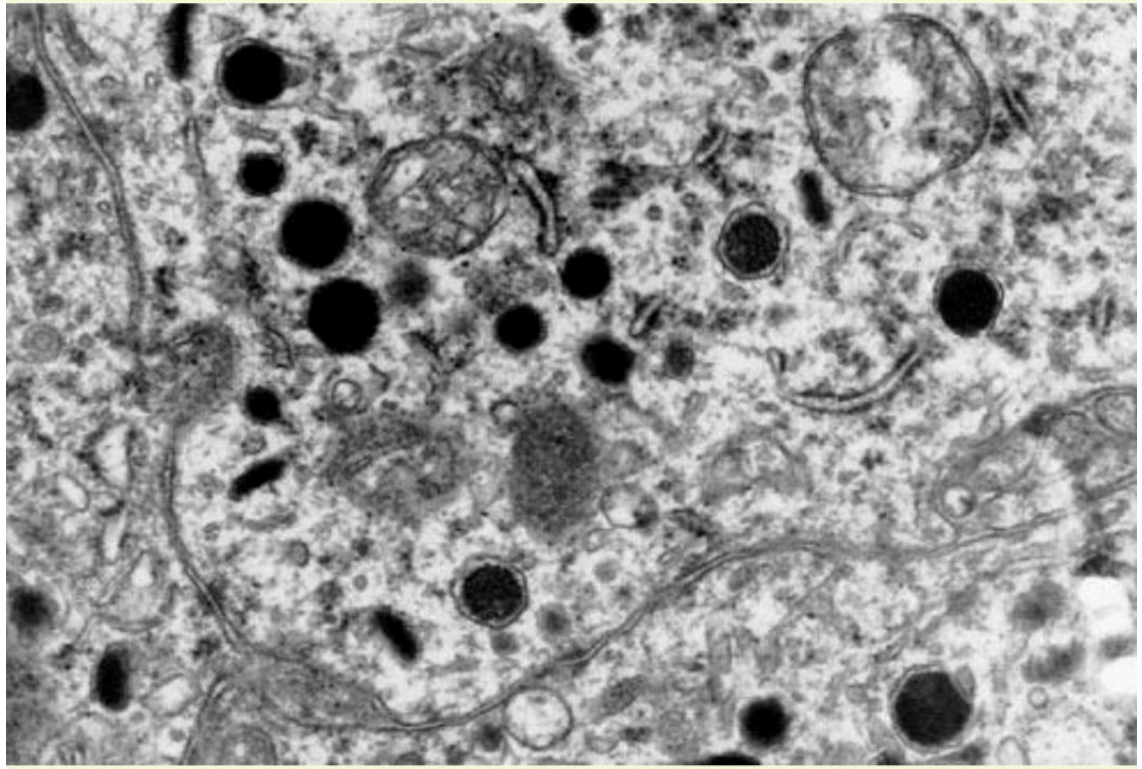
SLIDE # 118 - ADRENAL – PHEOCHROMOCYTOMA (20x)



HIGH-POWER VIEW OF THE TUMOR CELLS: NOTE THE MARKED VARIATION IN SIZE AND SHAPE

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PHEOCHROMOCYTOMA – ELECTRON MICROSCOPY IMAGE



MEMBRANE-BOUND DENSE CORE GRANULES IN THE CYTOPLASM

Adrenal vs. extra-adrenal pheochromocytomas Fill-in-the-blank

- _____% of pheochromocytomas are extra-adrenal
- Extra-adrenal pheochromocytomas are called _____
- _____, _____, and the bladder are extra-adrenal sites where these tumors may arise.
- Adrenal pheochromocytoma is _____ likely to be malignant than extra-adrenal tumors.

Adrenal vs. extra-adrenal pheochromocytomas Fill-in-the-blank

- 10 % of pheochromocytomas are extra-adrenal
- Extra-adrenal pheochromocytomas are called paraganglioma
- Organ of Zuckerkandl, Carotid body, and the bladder are extra-adrenal sites where these tumors may arise.
- Adrenal pheochromocytoma is less likely to be malignant than extra-adrenal tumors.



Name the familial tumor syndromes in which pheochromocytoma / paragangliomas arise.

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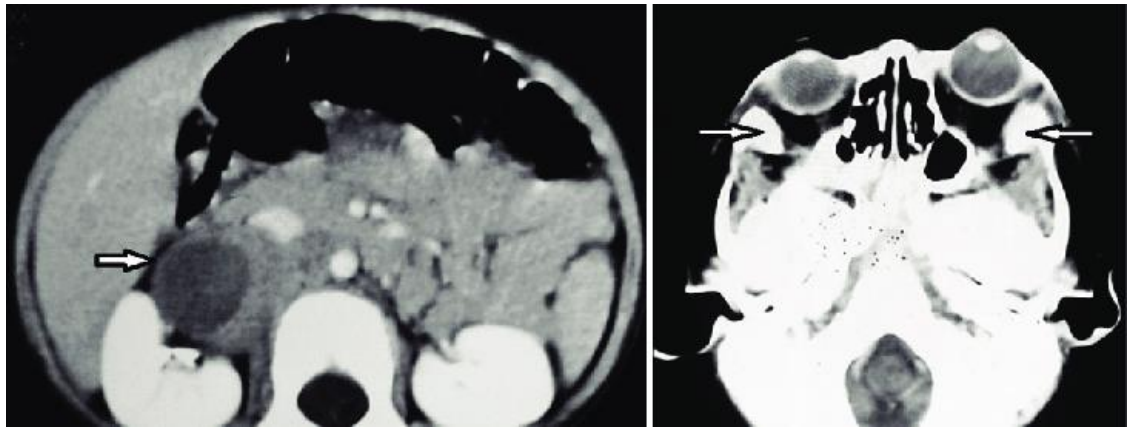
- Multiple endocrine neoplasia, type 2A (MEN2A)
- Multiple endocrine neoplasia, type 2B (MEN2B)
- Neurofibromatosis type 1 (NF1)
- Von Hippel Lindau (VHL)
- Familial paragangliomas

Case 4



A 3-year-old boy presents with a 2 month history of low-grade intermittent fever, abdominal pain, loss of appetite and weight loss. Bilateral periorbital ecchymoses developed over the past 2 weeks. Physical exam is significant for hepatosplenomegaly and a palpable abdominal mass.

Case 4 (cont.)

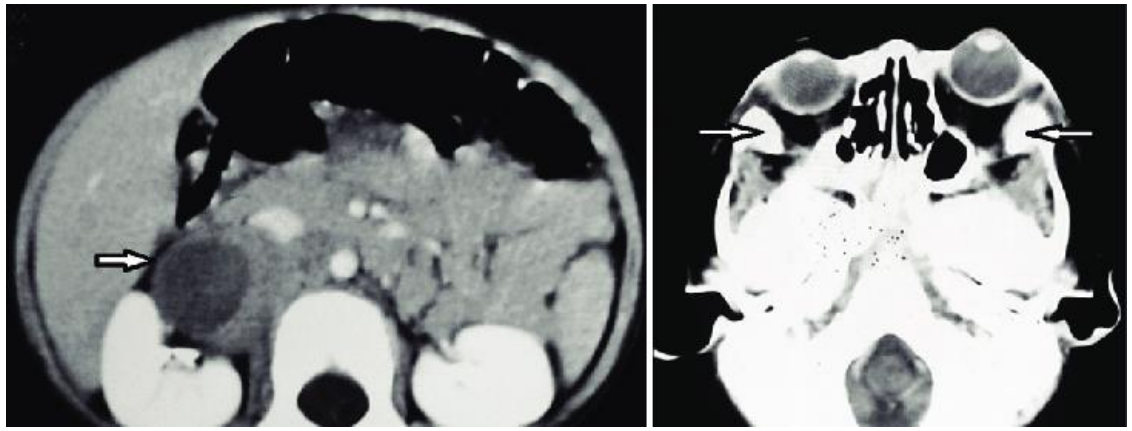


- CT abdomen showed a large right adrenal mass with solid and cystic components
- CT head/orbits showed extraconal hyperdense lesions in both orbits

Which of the following laboratory tests is most helpful for the diagnosis of this patient's tumor?

A. Flow cytometry
B. Mutation analysis for RET
C. Plasma ACTH
D. Plasma aldosterone
E. Urine vanillylmandelic acid (VMA) and homovanillic acid (HVA)

Answer: Case 4

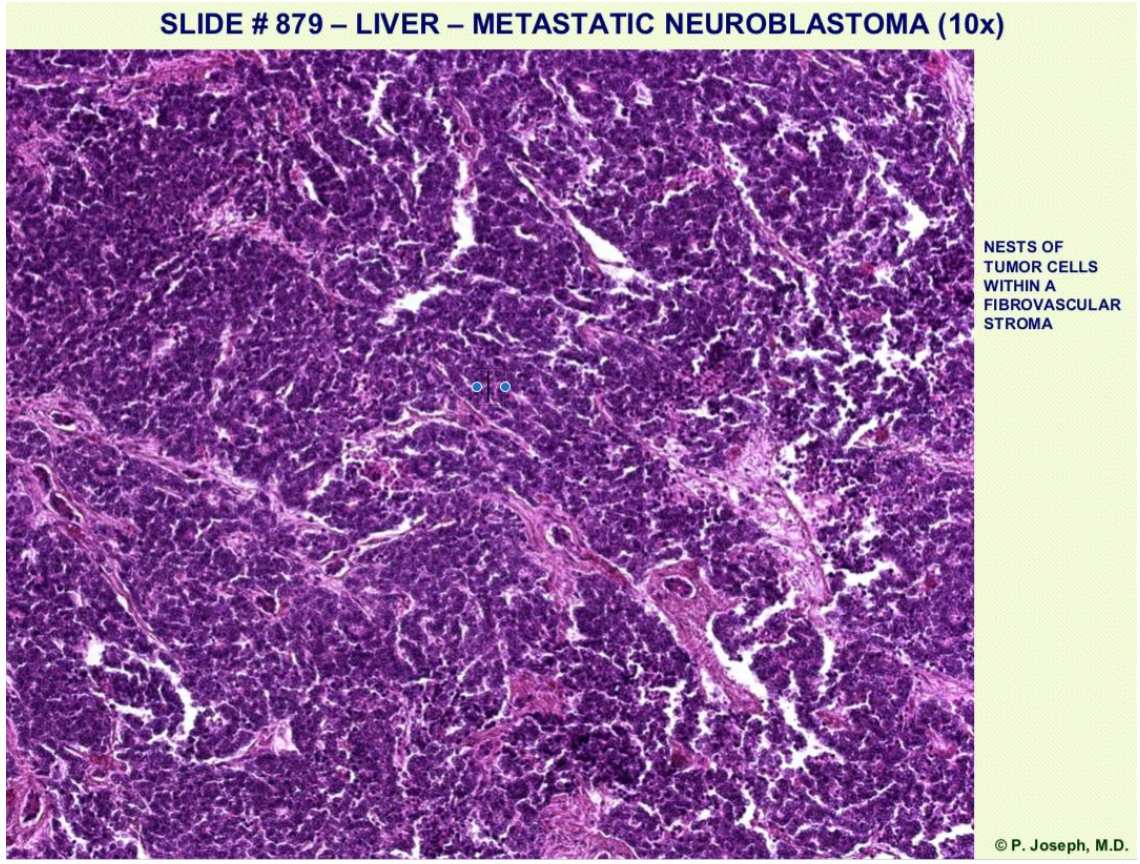


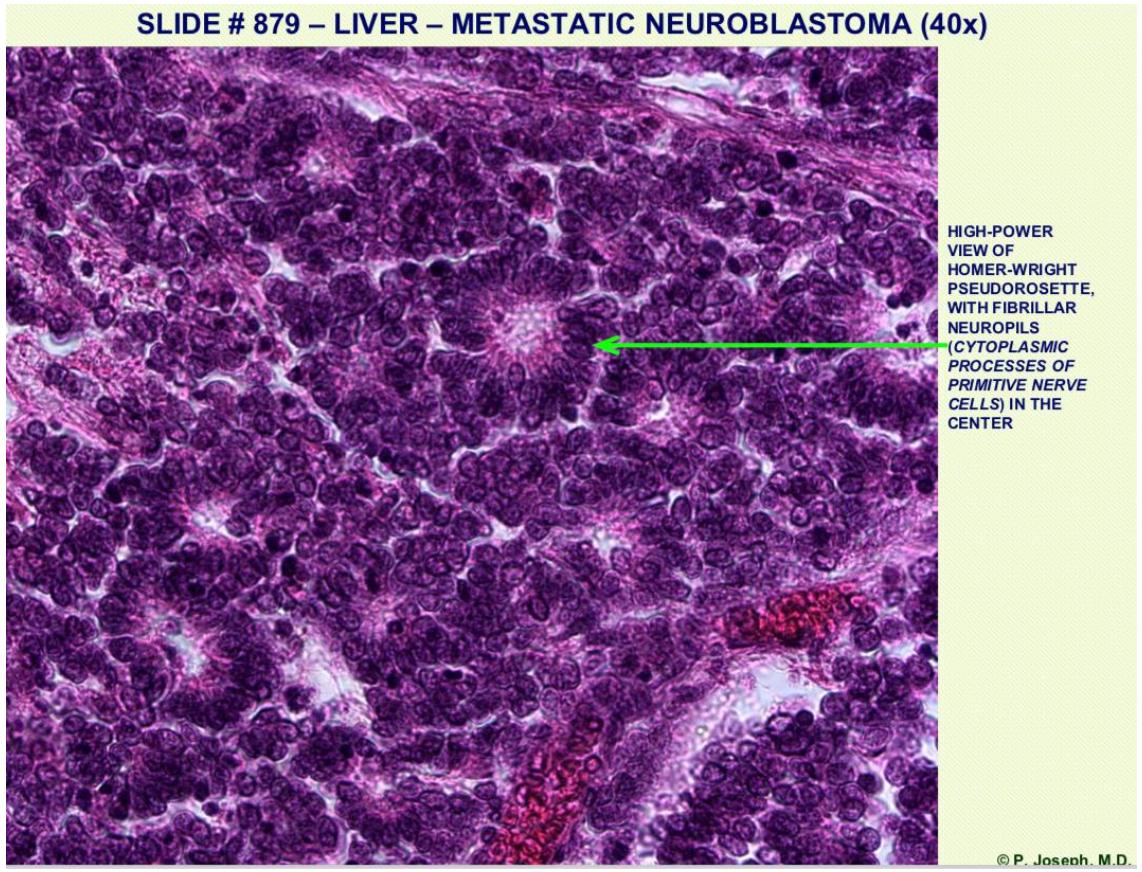
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Neuroblastoma
Microscopic features







Neuroblastoma

- What is the usual clinical presentation for these tumors?

- List some of the important prognostic factors in this tumor.

Neuroblastoma

- What is the usual clinical presentation for these tumors?
  1. **Neonates: disseminated neuroblastoma may present with multiple cutaneous metastases associated with deep blue discoloration of the skin**
  2. **<2 years: abdominal mass (usually painless), fever and weight loss**
  3. **Older children: clinical manifestations of metastatic disease, which may include hepatomegaly, ascites, and bone pain**

- List some of the important prognostic factors in this tumor.
  1. **Stage: low stage or 4S - favorable**
  2. **Age: <18 months - favorable**
  3. **Histology: Schwannian stroma or gangliocytic differentiation – favorable**
  4. **Genetics:**
     - **N-MYC amplification – unfavorable**
     - **Ploidy (chromosome count)**
       - **Hyperdiploid – favorable**
       - **Near diploid - unfavorable**

Case 5

A 27-year-old male presents for evaluation of headaches. On physical exam, his blood pressure is 140/85 mm Hg, and he has an enlarged thyroid. Laboratory studies show elevated serum calcitonin. A total thyroidectomy is performed. Sectioning reveals multiple tumor nodules in both lobes of the thyroid, composed of nests of neoplastic cells separated by amyloid-rich stroma. Further evaluation reveals the presence of an adrenal lesion.

What is the most likely accompanying morphologic change in the adrenal gland?

A. Bilateral lymphocytic infiltration with atrophy of all cortical zones
B. Bilateral cortical nodular hyperplasia
C. Solitary, 2cm, well-circumscribed cortical mass associated with atrophy of contralateral adrenal cortex
D. Large, 12 cm, yellow-tan mass with extensive necrosis arising in adrenal cortex
E. Bilateral, 2-4 cm masses arising in adrenal medullae

Answer: Case 5

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Question

For the patient in Case 5, the endocrinologist recommends genetic testing of the patient's family to determine risk for development of similar tumors.

Testing for a germline mutation of which of the following genes would be most helpful?

A. MEN1
B. RET
C. MYCN
D. NF1
E. PTEN

Answer

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A. MEN1
B. RET
C. MYCN
D. NF1
E. PTEN

Summary: MEN syndromes

	MEN1	MEN2A	MEN2B
Gene mutation	MEN1 gene	RET gene	RET gene
Tumors	**P**arathyroid hyperplasia and adenoma **P**ancreatic endocrine neoplasms (usually gastrinomas and insulinomas) **P**ituitary adenoma (usually prolactinoma)	Medullary carcinoma of thyroid Pheochromocytoma Parathyroid hyperplasia	Medullary carcinoma of thyroid Pheochromocytoma Extra-endocrine manifestations (ganglioneuromas of mucosal sites), marfanoid habitus

Part 2: Summary

- Pheochromocytoma
- Pheochromocytoma vs. paraganglioma
- Neuroblastoma
- MEN syndromes

Thank you!